B.TECH CAPSTONE PROJECT | FINAL VIVA VOCE EXAMINATION

Badminton Posture Detection and Injury Risk Assessment Using Machine Learning

An Automated, Non-Invasive Computer-Vision & Biomechanical Framework for Athletic Movement Screening

👤 Candidate Details

- **Candidate:** Pavan H S [Confirmed]
- **Degree:** Bachelor of Technology (B.Tech)
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- **Discipline:** Computer Science & Engineering [To Confirm]
- **Academic Year:** 2025 – 2026

🏛️ Academic Advisory

- **Internal Guide:** [Project Guide Name: To Confirm]
- **Head of Dept:** [HOD Name: To Confirm]
- **Department:** Dept. of Computer Science & Engineering
- **Institution:** [College / University Name: To Confirm]
- **Date:** October 2026

⚙️ System Summary

- **Pose Core:** Google MediaPipe (33 3D Landmarks)
- **Movement Model:** Random Forest (161 Reviewed Events)
- **Validation:** 3-Fold Player-Grouped CV (F1: 0.576, Acc: 0.621)
- **Risk Framework:** Rubric v0.3 (Provisional Observational)
- **Deployment:** Streamlit Dashboard + SQLite Audit Log

⚠️ Research Prototype Disclaimer: This system provides non-invasive biomechanical athletic screening based on sports science literature. It is an engineering prototype and does NOT provide certified clinical medical diagnoses or guarantee injury prevention.

BACKGROUND & CLINICAL NEED

The Biomechanical Challenge in High-Speed Racket Sports

Severe deceleration loads, high non-contact injury rates, and the accessibility barrier of modern motion analysis.

⚡ High-Deceleration Trauma

- **Extreme Impact Loading:** Deceleration forces during lunges and jump landings exceed 2.5–3.0× body weight.
- **Non-Contact Injuries:** High clinical incidence of ACL tears, patellar tendinopathy ('jumper's knee'), and ankle sprains.
- **Rapid Reaction Windows:** Ball speeds exceed 300 km/h; players have <300 ms reaction windows, leaving minimal margin for mechanical error.
- **Repetitive Strain:** Hundreds of explosive changes of direction per match cause cumulative joint fatigue.

🔬 Existing Lab Bottlenecks

- **Optical MoCap (Vicon/Qualisys):** Gold standard, but costs $50,000+, requires dedicated multi-camera rooms, and needs retro-reflective skin markers.
- **Wearable Sensors (IMUs):** Intrusive, alters natural racket swing mechanics, causes discomfort, and suffers from severe shock-induced sensor drift.
- **Grassroots Exclusion:** Elite biomechanical motion screening remains locked inside specialized research universities and Olympic centers.

💡 Computer Vision Opportunity

- **Monocular 2D Video:** Leverages standard lateral smartphone or court-side broadcast cameras without special hardware.
- **100% Non-Invasive:** Requires zero markers, zero wearable devices, and zero interference with athlete performance.
- **Automated Screening:** Extracts 3D pose, localizes critical plant keyframes, classifies movement actions, and flags mechanical risk indicators.
- **Accessible AI:** Brings elite sports medicine posture screening to club and academy players.

🔄 The Paradigm Shift in Athletic Motion Analysis

- **Traditional Paradigm:** Expensive Laboratory Marker MoCap ($50k+, restrictive markers, artificial lab environments, high setup overhead).
- **Proposed Vision AI Paradigm:** Accessible Monocular Pose Estimation + Biomechanical Machine Learning (markerless, natural match video, automated posture screening).

⚠️ Research Prototype — Observational Posture Screening (Not Clinical Diagnosis)

Technical Objectives & Explicit Scope Boundaries

Defining tangible engineering deliverables and maintaining strict scientific boundaries between CV and clinical medicine.

🎯 Core Engineering Objectives

- **1. Automated 3D Pose Extraction:** Track 33 anatomical landmarks from monocular video using Google MediaPipe Pose.
- **2. Kinematic Feature Pipeline:** Compute 10 physiologically validated sequence features (knee angles, torso lean, asymmetry, velocities).
- **3. Supervised Movement Classification:** Train an ML classifier on 161 human-reviewed sequences across three high-impact target classes.
- **4. Observational Risk Screening:** Implement Biomechanical Risk Rubric v0.3 to evaluate posture safety margins without diagnostic over-claims.
- **5. Interactive Web Dashboard:** Deliver an auditable Streamlit application with video playback, keyframe inspection, and local SQLite audit logging.

🛡️ Scope Boundaries & Non-Goals

- **Active Movement Scope:** Focus strictly on 3 high-impact classes: LUNGE, SMASH_LANDING, and CHANGE_OF_DIRECTION.
- **Deferred Action Classes:** SCISSOR_JUMP and FOOTWORK_SPLIT_STEP are retired from active scoring due to severe overlap and sample sparsity.
- **NOT a Clinical Predictor (Non-Goal):** Computer vision cannot measure bone density, ligament tensile strength, or muscle fatigue. The system evaluates POSTURE RISK INDICATORS, not medical outcomes.
- **Provisional Risk Framing:** Rubric v0.3 reports factual joint angles as 'PROVISIONAL' observations; discrete High/Moderate/Low labels are strictly disabled pending clinical trials.
- **Strict Safety Gating:** Low-quality or occluded frames automatically default to 'UNASSESSED' rather than hallucinating predictions.

⚠️ Crucial Distinction: Movement Classification vs. Injury Risk Assessment

- **Movement Classification (Machine Learning):** Determines WHAT movement occurred (e.g. Smash Landing vs Lunge) using Random Forest.
- **Injury Risk Assessment (Rule-Based Rubric v0.3):** Evaluates HOW that movement was executed biomechanically against sports science literature indicators.

SYSTEM ARCHITECTURE

Modular End-to-End Computer Vision Pipeline

A decoupled multi-stage architecture from raw video ingestion to auditable web presentation.

STAGE 1 Video Ingestion

- **OpenCV Decoded:** Raw MP4/MOV video streams.
- **Frame Resampling:** Subsampled at 15 FPS for efficiency.
- **Validation Guard:** Checks headers, corrupt frames, and resolution.
- **Frame Buffer:** Maintains sliding memory window.

STAGE 2 Pose Extraction

- **Google MediaPipe:** Pretrained Pose Landmarker.
- **33 3D Keypoints:** Tracks hips, knees, ankles, trunk, arms.
- **Quality Decoupling:** Separates lower-body from upper-limb quality.
- **Coordinate Norm:** Normalized image space + relative depth z.

STAGE 3 Candidate Detection

- **Temporal Segmentation:** Sliding-window kinematic detector.
- **Multi-Signal Cues:** Hip drop, stance width ratio, torso lean.
- **Keyframe Selection:** Local median smoothing identifies plant instant.
- **Subclip Slicing:** Extracts candidate event intervals.

STAGE 4 Feature & ML Engine

- **10 Kinematic Features:** Aggregated sequence statistics.
- **Quality Gating:** Rejects missing or non-finite measurements.
- **Random Forest:** 161-event reviewed model.
- **Multi-Class Probabilities:** Returns class confidence vector.

STAGE 5 Risk & Presentation

- **Rubric v0.3:** Evaluates knee flexion, trunk lean, asymmetry.
- **Provisional Status:** Reports factual degrees without over-claims.
- **Streamlit UI:** Annotated video + probability charts.
- **SQLite Audit Log:** Logs run history and JSON manifests.

🏛️ Decoupled Architectural Design Principle

- **Modular Decoupling:** Video I/O, Geometric Pose Tracking, Movement Classification, and Risk Rubrics operate as fully independent modules.
- **Regression Safety:** Extensively verified by our automated test suite (110 passed, 26 subtests passed across 12 test modules), ensuring zero silent regressions.

⚠️ Research Prototype — Observational Posture Screening (Not Clinical Diagnosis)

COMPUTER VISION & KINEMATICS

MediaPipe 3D Landmarks & Feature Engineering

Extracting 33 anatomical landmarks and condensing noisy coordinates into 10 physiologically grounded sequence features.

🦴 MediaPipe Pose & Quality Decoupling

- **Monocular 3D Landmarks:** MediaPipe extracts 33 landmarks with normalized (x, y) coordinates and relative depth z (depth relative to hip center).
- **The High-Speed Swing Problem:** Racket swings exceed 300 km/h, causing visual motion blur that degrades wrist and elbow confidence, dropping overall pose quality to ~0.12.
- **The Quality Decoupling Innovation:** Originally, overall quality < 0.40 rejected the entire video. We introduced an isolated Lower-Body Quality Metric (Q_lower).
- **Lower-Body Landmark Subset:** Evaluates hips (23, 24), knees (25, 26), and ankles (27, 28) independently.
- **The Result:** While an athlete's racket arm is blurred (overall quality 0.12), lower-body quality remains high (Q_lower >= 0.85). This prevents false pipeline rejections and safely enables knee/trunk posture evaluation!

📐 The 10 Anatomical Sequence Features

- **1. Lead Knee Flexion Min (θ_knee):** Deepest angle achieved during plant phase; indicates deceleration cushioning.
- **2. Trail Knee Flexion Min:** Assesses trailing limb stabilization and weight transfer.
- **3. Bilateral Knee Asymmetry:** |θ_left - θ_right|; flags unbalanced ground reaction force absorption.
- **4. Max Torso Forward Lean (θ_trunk):** Inclination of trunk vector relative to vertical; indicates lumbar strain.
- **5. Lateral Trunk Tilt:** Deviation from midline; core instability outside base of support.
- **6. Stance Width Ratio:** Inter-ankle distance divided by hip width; separates lunges from recovery steps.
- **7. Vertical Hip Displacement (Δy_hip):** Differentiates jump smash flight from ground footwork.
- **8. Peak Knee Flexion Velocity:** Angular rate of flexion; indicates landing shock severity.
- **9. Torso Angular Rate:** Rate of trunk tilt change during rapid directional cuts.
- **10. Stance Recovery Velocity:** Time rate of base return back to athletic ready position.
- **Vector Formulation:** Joint angle computed via 3D dot product: $\theta = \arccos((u \cdot v)/(|u||v|))$.

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TEMPORAL SEGMENTATION

Candidate Detection & Keyframe Localization

Isolating critical movement intervals and localizing peak mechanical plant depth.

🔍 Multi-Signal Candidate Detection

- **Continuous Sliding Window:** Buffers sequential video frames and computes running kinematic derivatives.
- **Vertical Hip Elevation (Δy_hip >= 0.04):** Triggers jump smash candidate intervals (takeoff, apex flight, and landing impact).
- **Stance Width Ratio (> 1.8x Hip Width):** Triggers forward lunge proposals by detecting wide base elongation during net recovery.
- **Lateral Acceleration & Trunk Lean (θ_trunk > 18°):** Triggers change-of-direction cuts when the player brakes sharply to reverse court travel.
- **Temporal Coalescing:** Merges overlapping detections into coherent event intervals (typically 0.8 to 2.5 seconds).
- **Subclip Extraction:** Slices candidate clips for human review and offline ML feature extraction.

🎯 Keyframe Selection (_select_representative_frame)

- **The Failure of Mathematical Midpoint:** Early prototypes selected the mathematical center of the clip (t_mid). In jump smashes, flight takes 75% of the clip and the landing occurs in the final 20%. Midpoint showed the player floating mid-air!
- **Our Plant-Depth Keyframe Algorithm:** Applies local median smoothing across knee flexion trajectories and hip elevations within the candidate window.
- **Peak Deceleration Localization:** Automatically isolates the exact split-second of maximum ground plant depth or landing impact.
- **Missing Pose Resilience:** Handles intermittent tracking dropouts gracefully without crashing or selecting corrupted frames.
- **Unit-Tested Robustness:** Verified across multiple test modules (test_candidate_detector.py, test_keyframe_selection.py) with 100% test pass rate.

ACTION TAXONOMY

Active Movement Classes & Class Refinement Rationale

Focusing on three high-risk athletic actions while pruning ambiguous, under-sampled classes.

🦵 LUNGE (Active)

- **Biomechanical Definition:** Deep forward or diagonal step with rapid knee deceleration to retrieve low net shots.
- **Clinical Importance:** Extreme eccentric quadriceps loading; high patellofemoral compressive force.
- **Key Identifiers:** Stance width > 1.8× hip width, lead knee flexion 90°–120°, forward trunk inclination.
- **Dataset Count:** 44 verified sequence events (27.3% of reviewed dataset).

🏸 SMASH_LANDING (Active)

- **Biomechanical Definition:** Ground impact deceleration following an overhead jump smash from the rear court.
- **Clinical Importance:** Peak ground reaction force (2.5–3.0× BW); dynamic knee valgus and landing shock.
- **Key Identifiers:** Vertical hip displacement Δy >= 0.04, rapid bilateral touchdown, high elbow cocking.
- **Dataset Count:** 72 verified sequence events (44.7% of reviewed dataset).

🔄 CHANGE_OF_DIRECTION (Active)

- **Biomechanical Definition:** Lateral push-off plant and aggressive directional cut to reverse court travel.
- **Clinical Importance:** Primary non-contact ACL injury mechanism; multi-planar knee shear loading.
- **Key Identifiers:** Lateral trunk lean outside base, high braking plant angle, rapid rebound acceleration.
- **Dataset Count:** 45 verified sequence events (28.0% of reviewed dataset).

⏸️ Retired & Deferred Classes: Scissor Jump & Footwork Split Step

- **SCISSOR_JUMP (Retired from active scoring):** Extreme kinematic overlap with changes of direction; high misclassification; sample sparsity in initial dataset (N < 5).
- **FOOTWORK_SPLIT_STEP (Retired from active scoring):** Preparatory low-amplitude bounce step with negligible injury risk; maintaining it diluted classifier precision on high-impact actions.
- **Engineering Rationale:** Pruning retired classes improved model precision on our 3 active high-risk classes without compromising system integrity.

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DATASET INTEGRITY

Dataset Staging Funnel & Rigorous Human Review

A transparent human-in-the-loop annotation pipeline with strict anti-data-pollution safeguards.

📊 The Staging Funnel

- **Raw Match Footage:** Recorded across 6 distinct players in natural court environments.
- **370 Automated Proposals:** Generated by multi-signal kinematic detection heuristics.
- **347 Subclips Sliced:** Segmented into discrete video files and indexed in review manifest.
- **161 Verified Sequence Events:** Validated through meticulous visual human review.
- **Reviewed Distribution:** • 72 Smash Landings (44.7%)
- 45 Changes of Direction (28.0%)
- 44 Lunges (27.3%)
- **Cohort Representation:** 6 distinct player groups (player_01 through player_06).

👁️ Human Review Schema

- **Strict Decision Triad:** Every candidate proposal must receive one of three explicit human decisions:
- **1. ACCEPT:** Visual confirmation unambiguously verifies target action kinematics (correct foot plant, jump takeoff, or lateral cut).
- **2. REJECT:** False detector trigger, camera occlusion, player stepping out of bounds, or background movement.
- **3. UNCERTAIN:** Ambiguous hybrid footwork, poor lighting, or transitional steps that cannot be verified.
- **Verification Integrity:** Automated heuristics were used ONLY to prioritize candidates for review, NEVER as proof of ground truth.

🛡️ Anti-Pollution Safeguards

- **Physical Data Separation:** Automated proposals and human review manifests are stored in separate directories.
- **Zero Synthetic Leakage:** Unreviewed candidate predictions are strictly forbidden from entering training datasets.
- **Immutable Baseline Data:** Original training datasets and historical model artifacts remain strictly preserved and hash-verified.
- **Auditable Extraction Script:** Training dataset generated via extract_reviewed_dataset.py, requiring explicit human decision flags.
- **Scientific Credibility:** Ensures the machine learning model learns from validated biomechanics rather than its own detector artifacts.

EMPIRICAL EVALUATION

Model Evaluation: 3-Fold Player-Grouped Cross-Validation

Evaluating realistic cross-subject generalization without identity leakage.

MACRO-F1 SCORE
0.5760
(57.6%) Balanced Class Avg

GROUPED ACCURACY
0.6211
(62.1%) Cross-Subject Metric

SMASH LANDING F1
0.819
Prec: 0.778 | Rec: 0.865

LUNGE & COD F1
0.500 / 0.409
Lunge Rec: 0.455 | COD: 0.432

Evaluation Metric	Baseline	Random Forest	Improvement
Accuracy	0.329 (32.9%)	0.621 (62.1%)	+29.2% (2x)
Macro-F1	0.354 (35.4%)	0.576 (57.6%)	+22.2% Lift
SMASH_LANDING F1	0.582	0.819	+0.237 Lift
LUNGE / COD F1	0.240 / 0.241	0.500 / 0.409	+0.260 / +0.168

🔬 GroupKFold & Cohort Analysis

- **The Data Leakage Fallacy:** Standard random train/test splits leak player identity (same clothing, lighting, body mass), claiming false >90% accuracy.
- **Strict Player Holdout:** In 3-fold GroupKFold, test fold players were NEVER seen during training. Measures true cross-subject transfer.
- **Cohort Imbalance Limitation:** Player 05 accounts for 48% of events. Holding out Player 05 drops Fold 1 F1 to 0.485, whereas Fold 3 reaches 0.702.
- **Smash vs Lunge/COD Disparity:** Smash Landing achieves high F1 (0.819) due to vertical hip displacement. Lunge recall (0.455) and COD F1 (0.409) suffer because lateral cuts and short lunges share similar knee flexion angles in 2D video.
- **Scientific Baseline:** 62.1% accuracy and 0.576 macro-F1 represent an honest, validated cross-subject baseline—not guaranteed deployment performance.

BIOMECHANICAL RISK FRAMEWORK

Rubric v0.3: Observational Risk Screening vs. Clinical ML

Why rule-based biomechanical rubrics are used instead of pseudo-scientific trained ML models.

⚖️ Why Rubric v0.3 Instead of Trained Clinical ML?

- **The Clinical Ground Truth Reality:** To train a supervised ML model on 'injury risk', you need thousands of prospective athlete videos paired with MRI-confirmed clinical injury records (e.g. ACL tears within 30 days).
- **The Pseudo-Science Risk:** No such clinical dataset exists in computer vision literature. Training an ML model on synthetic 'High/Low' labels creates dangerous machine learning hallucination.
- **Our Principled Solution:** Use Machine Learning strictly for what vision can verify—Movement Classification—and use published sports medicine biomechanical rubrics for risk screening.
- **Evidence-Based Literature:** Rubric v0.3 indicators are grounded in published sports medicine studies (Koga et al., Hewett et al.) on ACL injury mechanisms.

📋 Rubric v0.3 Assessment States

- **1. PROVISIONAL (Unvalidated Cutoffs):** Assigned when tracking quality is verified and movement is classified as LUNGE, SMASH_LANDING, or CHANGE_OF_DIRECTION.
- **Factual Joint Metrics Reported:** Reports measured lead knee flexion, trunk forward lean, lateral tilt, and bilateral asymmetry in degrees without making diagnostic claims.
- **Published Safety Boundaries:** Screens against literature indicators: stiff knee flexion (<90°), excessive torso lean (>25°), and asymmetry (>30°).
- **2. UNASSESSED Status:** Safely triggered when lower-body tracking quality drops (Q_lower < 0.40), required landmarks are occluded, or movement is UNCERTAIN.
- **Fail-Safe Safety Architecture:** Refuses to guess or output false reassurance when visual data is insufficient.

⚠️ Rubric v0.3 Clinical Policy: Discrete HIGH / MODERATE / LOW Scoring is Disabled

- **Ethical Policy:** Under Rubric v0.3, discrete 'HIGH', 'MODERATE', and 'LOW' risk categorization is strictly disabled pending multi-center empirical biomechanical trials.
- **Scientific Stance:** Monocular camera angles introduce 2D perspective foreshortening; reporting uncalibrated categorical risk would be clinically irresponsible.

SYSTEM VALIDATION

Interactive Dashboard Architecture & Verified Demo Cases

Real-time web inference, automated keyframe rendering, and fail-safe uncertainty handling.

🏸 Case 1: Overhead Smash

- **Video Clip:** test_smash_movement1.mp4 (3.88s, 1080p)
- **Tracking Quality:** Overall: 0.621 | Lower-Body: 0.884
- **Movement Output:** SMASH_LANDING (65.9% confidence)
- **Candidate Detector:** Proposed CHANGE_OF_DIRECTION; RF correctly classified as SMASH_LANDING.
- **Observed Biomechanics:** Lead knee flexion: 102.1°, Torso forward lean: 14.7°, Asymmetry: 27.7°.
- **Risk Status:** PROVISIONAL (Unvalidated Cutoffs)

🦵 Case 2: Forward Net Lunge

- **Video Clip:** cand_006_LUNGE...mp4 (1.27s)
- **Tracking Quality:** Overall: 0.937 | Lower-Body: 0.951
- **Movement Output:** LUNGE (94.1% confidence)
- **Keyframe Selection:** Isolated frame 23 (0.44s), capturing exact peak plant depth.
- **Observed Biomechanics:** Lead knee: 102.5°, Asymmetry: 8.3° (minimal bilateral asymmetry).
- **Risk Status:** PROVISIONAL (Unvalidated Cutoffs)

❓ Case 3: Ambiguity Safeguard

- **Video Clip:** cand_014_UNCERTAIN...mp4 (1.47s)
- **Candidate Detector:** UNCERTAIN (No movement heuristic met threshold).
- **Classifier Suggestion:** CHANGE_OF_DIRECTION (63.6%)
- **Safeguard Triggered:** System REFUSES to override detector uncertainty. Final label remains UNCERTAIN.
- **Risk Status:** UNASSESSED (Safely aborted)
- **Scientific Value:** Guarantees zero false predictions on ambiguous footwork.

💻 Application Architecture on Port 18502

- **Interactive Streamlit Engine:** Live video uploader, annotated playback with skeletal overlay, keyframe inspection, and probability charts.
- **SQLite Run History Database:** Persistent session logging of video names, timestamps, joint angles, and JSON manifests for longitudinal tracking.

REFLECTIONS & ROADMAP

Technical Limitations, Future Priorities & Conclusion

Reflecting on research constraints, outlining next-phase engineering, and project wrap-up.

⚠️ Technical Limitations

- **Monocular Perspective Error:** 2D camera foreshortening distorts 3D joint angle measurements during off-axis court movements.
- **Anatomical Self-Occlusion:** Deep lunges away from camera cause the lead thigh to occlude the trailing leg.
- **Small Cohort Constraint:** Evaluated across 6 athletes with Player 05 representing 48% of events; limits cross-demographic transfer.
- **Motion Blur at 30 FPS:** High-speed racket swings blur wrists and rackets, requiring lower-body quality decoupling.

🚀 Future Research Roadmap

- **1. Multi-Camera Synchronized Capture:** Calibrate multi-view stereo cameras to eliminate monocular foreshortening and reconstruct metric 3D MoCap.
- **2. High-Frame-Rate Capture (120–240 FPS):** Eliminates high-speed racket swing blur and provides sub-frame landing impact resolution.
- **3. Prospective Sports Medicine Study:** Collaborate with sports biomechanics clinics to establish calibrated injury thresholds from longitudinal athlete health records.
- **4. Expanded Demographic Cohorts:** Expand dataset to 50+ athletes across gender, age, and competitive playing tiers.

🏁 Project Conclusion

- **End-to-End Vision Pipeline:** Successfully designed, built, and verified a complete monocular AI framework for badminton motion analysis.
- **Rigorous ML Evaluation:** Achieved 62.1% accuracy and 0.576 macro-F1 under zero-leakage Player-Grouped Cross-Validation (+29% over baseline).
- **Biomechanical Innovation:** Decoupled lower-body tracking quality and developed plant-depth keyframe localization.
- **Scientific Integrity:** Maintained strict separation between ML movement classification and observational risk screening.
- **Production-Grade Engineering:** Streamlit UI, SQLite audit logging, and 110 automated tests passing with 100% pass rate.